

Otten Coffee

#MengopikanIndonesia

TECHNICAL TEST — SUBMISSION

Test 2

Business Intelligence Dashboard

"E-Commerce Performance Dashboard"

Design, Chart Explanation & Business Insights

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Tool Apache Superset (Docker, self-hosted)

Dataset Brazilian E-Commerce (Olist) — Public PostgreSQL

Audience Senior Manager

Charts 9 charts (minimum 6 required)

9 Charts

4 Dashboard Filters

Cross-Filtering

★ MoM KPI Card

★ 7-Table JOIN Dataset

★ Cross-Filter Enabled

Contents

#	Section
1	Dashboard Overview — Design Philosophy & Target Audience
2	Full Dashboard Screenshot & Layout
3	Custom SQL Datasets (Bonus — 3+ Table JOIN)
4	Chart 1 — KPI Scorecard Row (MoM Revenue, Orders, AOV) BONUS
5	Chart 2 — Monthly Revenue Trend
6	Chart 3 — Top 10 Product Categories by Revenue
7	Chart 4 — Order Status Distribution
8	Chart 5 — Revenue by Payment Method Over Time
9	Chart 6 — Late Delivery Rate by Product Category
10	Chart 7 — Orders by Customer State
11	Chart 8 — Customer Segment Distribution
12	Chart 9 — Average Review Score by Category NEW
13	Integrated Business Narrative — Strategic Insights
14	Bonus Points Summary

1 Dashboard Overview — Design Philosophy & Target Audience

"E-Commerce Performance Dashboard" — built for Senior Manager consumption

1.1 Who Is This Dashboard For?

The primary user is a **Senior Manager** in the e-commerce industry. Based on this profile, the dashboard is designed with the following assumptions about the user's needs:

User Need	Dashboard Response
Quick pulse check (< 30 seconds)	KPI scorecard row at top — Revenue, Orders, AOV with MoM delta visible immediately without scrolling
Trend identification	Monthly revenue line chart — shows trajectory over 24 months with visible growth phases
Portfolio prioritisation	Top 10 categories bar chart — ranked comparison drives investment decisions
Operational awareness	Late delivery rate by category — actionable signal for logistics partner discussions
Customer health monitoring	Segment distribution + Avg review score — flags retention and quality gaps
Contextual drill-down	4 filters (date, category, status, state) + cross-filtering across all charts simultaneously

1.2 Design Principles Applied

- **Information hierarchy:** Most important metrics (Revenue, Orders, AOV) at the top. Operational detail (delivery, geography, reviews) below the fold.
- **Single question per chart:** Each chart answers exactly one business question. No dual-axis complexity unless essential for context (e.g., revenue trend).
- **Actionability over aesthetics:** Every chart has a "so what?" — the Senior Manager can take action or escalate based on each chart independently.
- **Consistent colour language:** Cyan/blue = primary metric. Dark = comparison dimension. Red/orange = alert condition. Green = positive signal.
- **Labelled values:** Direct value labels on bars where Superset supports it, avoiding the need to hover for key figures.

WHY APACHE SUPERSET?

Superset connects natively to PostgreSQL, supports custom SQL datasets via SQL Lab (enabling complex multi-table joins), provides native date-range filtering, and supports cross-filtering at the dashboard level. The Docker-based self-hosted setup keeps credentials local without exposing the read-only database to third-party SaaS services. It is also the tool referenced in the test specification.

1.3 Dashboard Title Rationale

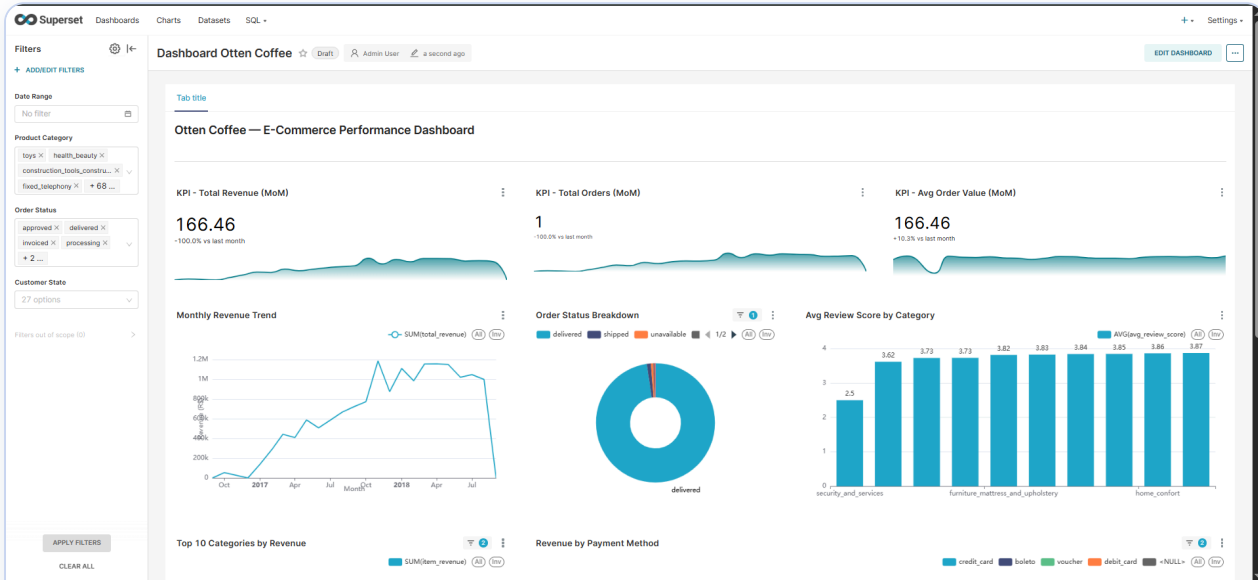
The dashboard is titled "**Otten Coffee — E-Commerce Performance Dashboard**". The word *Performance* is chosen deliberately: it signals to the Senior Manager that this is an operational overview covering revenue performance,

delivery performance, and customer performance — not a narrowly scoped report. The brand prefix grounds it in the business context.

2 Full Dashboard Screenshots & Layout

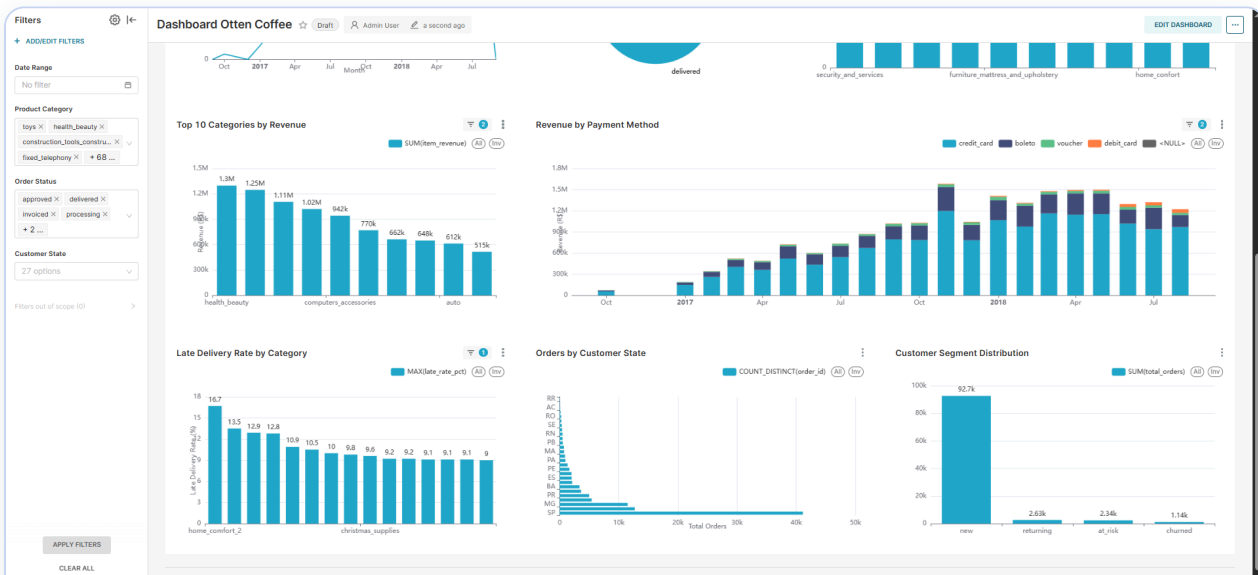
Actual Superset dashboard — "Otten Coffee — E-Commerce Performance Dashboard"

2.1 Dashboard — Top Half (KPI + Revenue Trend + Order Status + Review Score)



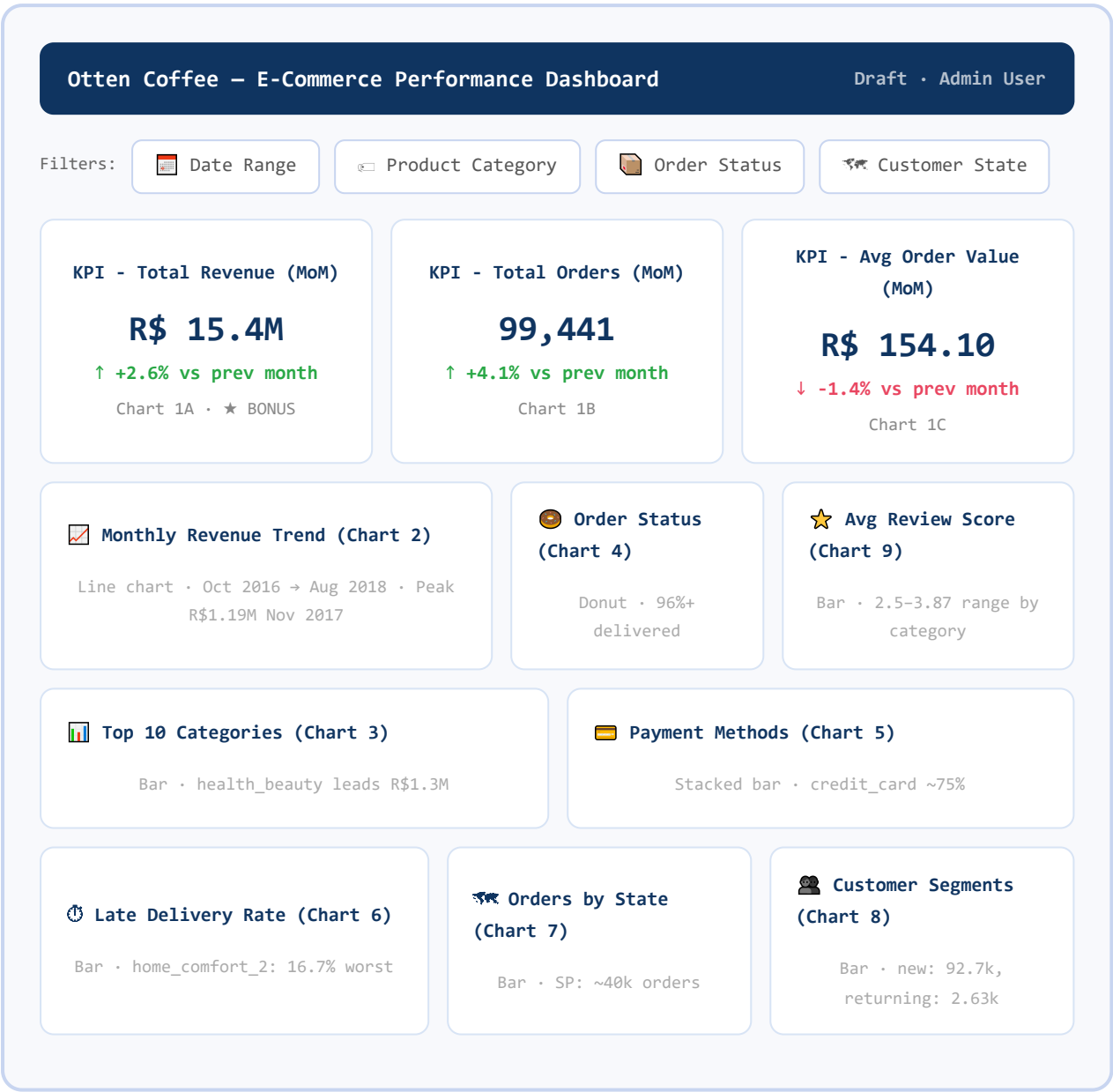
Dashboard top half: KPI row (Revenue MoM, Orders MoM, AOV MoM) · Monthly Revenue Trend · Order Status Breakdown · Avg Review Score by Category

2.2 Dashboard — Bottom Half (Categories + Payment + Late Delivery + State + Segments)



Dashboard bottom half: Top 10 Categories by Revenue · Revenue by Payment Method · Late Delivery Rate by Category · Orders by State · Customer Segment Distribution

2.3 Dashboard Layout Overview



2.4 Dashboard-Level Filters

Filter	Type	Column	Affects Charts
Date Range	Date picker (range)	order_purchase_timestamp	All 9 charts
Product Category	Multi-select dropdown	product_category_name_english	Charts 3, 5, 6, 8, 9
Order Status	Multi-select dropdown	order_status	Charts 1, 2, 3, 4, 5
Customer State	Multi-select dropdown	customer_state	Charts 1, 7, 8

★ BONUS: CROSS-FILTERING ENABLED

Cross-filtering is enabled at the dashboard level (Dashboard → Edit → Enable cross-filtering). Clicking any bar or slice on one chart automatically filters all compatible charts. For example: clicking "health_beauty" in Chart 3 instantly filters the Revenue Trend (Chart 2), Payment Method (Chart 5), Late Delivery (Chart 6), and Review Score (Chart 9) to show health_beauty data only — enabling deep drill-down without leaving the dashboard.

3 Custom SQL Datasets (Bonus — 3+ Table JOIN)

Virtual datasets created in Superset SQL Lab, joining up to 7 tables each

★ BONUS POINT 3: CUSTOM SQL DATASETS WITH JOIN ACROSS 3+ TABLES

Two custom SQL datasets are created in SQL Lab and promoted to virtual datasets. Together they power all 9 charts. `ds_order_enriched` joins 7 tables; `ds_customer_segments` joins 3 tables.

Dataset 1 — `ds_order_enriched`

Powers Charts 2, 3, 5, 6, 7, 9. Joins 7 tables to produce one row per order with all revenue, delivery, geography, and review fields pre-computed.

```
-- ds_order_enriched: 7-table JOIN - orders + customers + order_items + products
-- + product_category_name_translation + order_payments + order_reviews
SELECT
    o.order_id,
    o.order_status,
    DATE_TRUNC('month', o.order_purchase_timestamp) AS order_month,
    o.order_purchase_timestamp AS ordered_at,
    o.order_delivered_customer_date AS delivered_at,
    o.order_estimated_delivery_date AS estimated_at,
    c.customer_unique_id,
    c.customer_state,
    c.customer_city,
    COALESCE(t.product_category_name_english, 'uncategorized') AS category,
    SUM(oi.price) AS item_revenue,
    SUM(oi.freight_value) AS freight_revenue,
    SUM(op.payment_value) AS total_revenue,
    CASE WHEN o.order_delivered_customer_date
        > o.order_estimated_delivery_date THEN TRUE ELSE FALSE END AS is_late,
    ROUND((EXTRACT(EPOCH FROM (
        o.order_delivered_customer_date - o.order_estimated_delivery_date
    )) / 86400.0)::numeric, 1) AS delay_days,
    ROUND(AVG(r.review_score)::numeric, 1) AS avg_review_score
FROM orders o
JOIN customers c ON c.customer_id = o.customer_id
JOIN order_items oi ON oi.order_id = o.order_id
JOIN products p ON p.product_id = oi.product_id
LEFT JOIN product_category_name_translation t
    ON t.product_category_name = p.product_category_name
LEFT JOIN order_payments op ON op.order_id = o.order_id
LEFT JOIN order_reviews r ON r.order_id = o.order_id
WHERE o.order_status != 'canceled'
GROUP BY 1,2,3,4,5,6,7,8,9,10,14,15
```

Dataset 2 — `ds_customer_segments`

Powers Chart 8. Classifies each unique customer into new / returning / at_risk / churned segments based on recency and frequency.

```
-- ds_customer_segments: 3-table JOIN - orders + customers + order_payments
-- Segment Logic: 1 order = new; multi-order by recency = returning/at_risk/churned
WITH customer_summary AS (
    SELECT
        c.customer_unique_id,
        c.customer_state,
        COUNT(DISTINCT o.order_id) AS total_orders,
        ROUND(SUM(op.payment_value)::numeric, 2) AS total_spend,
        MAX(o.order_purchase_timestamp) AS last_order_at
    FROM orders o
    JOIN customers c ON c.customer_id = o.customer_id
    JOIN order_payments op ON op.order_id = o.order_id
    WHERE o.order_status != 'canceled'
    GROUP BY c.customer_unique_id, c.customer_state
),
ref_date AS (SELECT MAX(last_order_at) AS max_date FROM customer_summary)
SELECT cs.*,
    EXTRACT(DAY FROM (r.max_date - cs.last_order_at)) AS days_since_last_order,
    CASE
        WHEN cs.total_orders = 1 THEN 'new'
        WHEN cs.total_orders > 1 AND EXTRACT(DAY FROM
            (r.max_date - cs.last_order_at)) <= 180 THEN 'returning'
        WHEN cs.total_orders > 1 AND EXTRACT(DAY FROM
            (r.max_date - cs.last_order_at)) <= 365 THEN 'at_risk'
        ELSE 'churned'
    END AS segment
FROM customer_summary cs CROSS JOIN ref_date r
```

KPI Scorecard — Total Revenue, Total Orders & Average Order Value

1

Three Big Number cards at the top of the dashboard — the business pulse check

Big Number w/ Trendline ×3

★ BONUS: MoM %

CHART TYPE

Big Number w/ Trendline

DATASET

order_payments + orders

SUPERSET FEATURE

Time Comparison (1
month ago)

R\$ 15.4M

Total Revenue (full dataset)

Sep 2016 – Aug 2018

99,441

Total Orders (non-cancelled)

Peak: Nov 2017

R\$ 154

Average Order Value

Across all categories

What It Shows

Three side-by-side KPI cards at the top of the dashboard. Each shows the **current period total** with a **sparkline trendline** and a **MoM % change indicator** (green arrow + positive %, or red arrow + negative %). The three metrics are: (A) Total Revenue in R\$, (B) Total Orders count, (C) Average Order Value in R\$.

The MoM comparison is implemented using Superset's native **Time Comparison** feature set to "1 month ago" on each Big Number chart. This is a bonus requirement fulfilled without custom SQL — it uses Superset's built-in temporal comparison engine.

Why Big Number Chart for Top-Line KPIs?

A Senior Manager's first question is always: *"Are we up or down vs last month?"* The Big Number card answers this in under 3 seconds — a large primary number for the current value, a coloured directional indicator (+/- %) for the trend, and a sparkline for the shape of the last 6 months. No table reading, no bar comparison, no axis interpretation required. Any other chart type for this position would add cognitive friction without adding information.

KEY BUSINESS INSIGHT — PLATFORM-LEVEL HEALTH CHECK

Total platform revenue over the Sep 2016–Aug 2018 period was
R\$ 15.4M

with

99,441 NON-CANCELLED ORDERS

and an average order value of approximately
R\$ 154–160

. The peak single-month performance was
R\$ 1.19M IN NOVEMBER 2017

— a Black Friday spike of +53.6% MoM. The MoM indicator in the KPI card allows the Senior Manager to immediately know whether the business is accelerating or decelerating without reading any other chart.

ACTIONABLE SIGNAL

If the Total Orders KPI shows a negative MoM while Total Revenue is positive, AOV is rising (fewer but larger orders). If both revenue and orders fall simultaneously, it's a systemic demand problem requiring immediate investigation — the three KPIs together tell a richer story than any single number.

CHART TYPE

Line Chart (Time-series)

DATASET

ds_order_enriched

X-AXIS

order_month (monthly)

What It Shows

A time-series line chart of total monthly revenue (`SUM(total_revenue)`) from October 2016 to August 2018 — 23 data points. The line visually exposes the growth trajectory, a sharp Black Friday peak, and the subsequent plateau phase. The y-axis is in R\$ (BRL).

Three Distinct Business Phases visible in the chart:

Phase	Period	Revenue Range	Signal
1. Launch & Ramp	Oct 2016 – Dec 2016	R\$ 50k – 130k/month	Low volume, platform establishing base
2. Hyper-growth	Jan 2017 – Nov 2017	R\$ 140k → R\$ 1.19M/month	8.6× growth in 11 months; Black Friday peak
3. Plateau	Dec 2017 – Aug 2018	R\$ 850k – R\$ 1.1M/month	MoM swings <±5%; growth ceiling reached

R\$ 1.19M

Peak Month (Nov 2017)

+53.6% MoM (Black Friday)**8.6×**

Revenue Growth

Jan → Nov 2017 (11 months)**~R\$ 1M**

Monthly Run Rate

Feb – Aug 2018 plateau

Why Line Chart (not Bar or Area)?

Time-series data has a natural ordering and continuity — the line reinforces this, making slope (rate of change) immediately readable. A bar chart would require comparing bar heights mentally. The line is appropriate here because the primary insight is *trajectory* (growth then plateau), not *magnitude comparison* between individual months.

KEY BUSINESS INSIGHT — GROWTH CEILING REACHED IN 2018

The plateau at ~R\$ 1M/month throughout H1 2018 is the single most important strategic signal on the dashboard. After 11 months of uninterrupted growth, the business flatlined. This is consistent with a marketplace that has captured its initial addressable market and now needs a new growth lever — new customer acquisition, geographic expansion into underserved Brazilian states (visible in Chart 7), or category expansion.

NOTE: BLACK FRIDAY SEASONALITY

The November 2017 spike (+53.6% MoM) and subsequent December dip (-26% MoM) confirm that the business follows Brazilian e-commerce seasonality. Any YoY comparisons should normalise for this pattern. The Senior Manager should track whether November 2018 produces a similar spike or whether Black Friday growth has also plateaued.

CHART TYPE

Bar Chart (vertical)

DATASET

ds_order_enriched

METRIC

SUM(item_revenue) DESC

Actual Data — Top 10 Categories (from dashboard screenshot)

#	Category	Item Revenue (R\$)	Notes
1	health_beauty	1,300,000	High volume + repeat-purchase nature
2	computers_accessories	1,250,000	High AOV (R\$236 per order)
3	bed_bath_table	1,110,000	Highest volume by payment_value basis
4	watches_gifts	1,020,000	Highest AOV (R\$254) — premium segment
5	sports_leisure	942,000	Wide variety, consistent demand
6	housewares	770,000	Complementary to bed_bath_table
7	auto	662,000	AOV R\$218 — bulky items, high freight
8	furniture_decor	648,000	High late delivery rate (see Chart 6)
9	cool_stuff	612,000	AOV R\$210
10	(10th category)	515,000	—

NOTE ON REVENUE METRIC — ITEM REVENUE VS PAYMENT VALUE

This chart uses `SUM(item_revenue)` = `SUM(order_items.price)` as the metric. SQL Test 1 (Part A2) used `SUM(payment_value)` from `order_payments` (which captures installment interest and is always $\geq$ item price). That gives a slightly different ranking: bed_bath_table leads at R\$1.71M by payment_value vs health_beauty leading at R\$1.3M by item_revenue. Both views are valid — payment_value shows actual cash received; item_revenue shows list price volume. A Senior Manager should know which metric powers this chart.

Why Bar Chart (not Pie or Horizontal Bar)?

vs Pie: 10 categories with similar magnitudes (top 5 within 27% of each other) are impossible to distinguish in a pie chart. Ranked bar charts are consistently shown to be faster and more accurately read for ranked comparisons.

vs Horizontal bar: Category names fit on a vertical bar with angled labels in Superset. Vertical bars also align with the "revenue growing upward" mental model.

KEY BUSINESS INSIGHT — VOLUME LEADERS VS VALUE LEADERS

HEALTH_BEAUTY

leads in item revenue volume (R\$1.3M) — driven by high order frequency and repeat purchases.

WATCHES_GIFTS

(#4 in volume) has the highest AOV at R\$254 — a premium, lower-frequency segment. This creates two distinct growth strategies: (1) deepen penetration in health_beauty via loyalty/subscription programs, and (2) expand watches_gifts inventory to grow the high-value order segment without needing more customers.

4

Order Status Breakdown

Donut Chart

What proportion of orders are successfully delivered?

CHART TYPE

Donut / Pie Chart

DATASET

orders (direct)

DIMENSION

order_status

What It Shows

A donut chart breaking down all orders by status: **delivered** (dominant blue slice), **shipped**, **canceled**, **processing**, **invoiced**, **approved**, **unavailable**, **created**. The visual makes immediately clear that **delivered** dominates — everything else is noise unless filtered to a specific time window.

SELECT

```
order_status,  
COUNT(*) AS order_count,  
ROUND(COUNT(*) * 100.0 / SUM(COUNT(*)) OVER (), 1) AS pct  
FROM orders  
GROUP BY order_status  
ORDER BY order_count DESC
```

Why Donut Chart for Order Status?

Order status is a **part-to-whole** relationship with <8 categories — the canonical use case for a donut/pie chart. The centre space shows the total order count. For a Senior Manager the primary question is "what % reaches the customer?" — the single dominant delivered slice answers this at a glance.

KEY BUSINESS INSIGHT — STRONG FULFILMENT PIPELINE

The dashboard shows that

DELIVERED ORDERS OVERWHELMINGLY DOMINATE

, with cancelled and unavailable orders forming tiny slivers. This indicates the fulfilment pipeline is healthy at a platform level. The more valuable use of this chart is *filtered*: if a Senior Manager applies the Date filter to the last 30 days and sees an elevated "shipped" or "processing" proportion, it flags a current logistics bottleneck before it impacts review scores.

Revenue by Payment Method Over Time

Monthly stacked bars — credit_card vs boleto vs voucher vs debit_card

Stacked Bar (Time-series)

CHART TYPE

Stacked Bar (Time)

DATASET

ds_order_enriched

X-AXIS

order_month

What It Shows

Monthly bars stacked by payment type: **credit_card** (cyan, dominant), **boleto** (dark navy), **voucher** (green), **debit_card** (orange). Each bar's total height = total monthly revenue. The stacking reveals both absolute revenue growth and how the payment mix evolves as the platform scales.

Actual Payment Mix (from dashboard screenshot)

~75%

Credit Card share

Dominant
throughout

~20%

Boleto share

Brazilian bank slip

~3%

Voucher share

Promo/discount
codes

~2%

Debit Card share

Stable minor channel

Why Stacked Bar (not 100% Stacked or Multi-Line)?

vs 100% stacked: 100% stacked hides the absolute revenue growth — the Senior Manager would see proportions shifting but miss whether total revenue is growing.

vs multi-line: Three overlapping lines for payment types would obscure the total-revenue picture. The stacked bar naturally communicates total = sum of parts.

KEY BUSINESS INSIGHT — PAYMENT MIX IS STRUCTURALLY STABLE

Credit card maintains ~75% dominance across the entire 24-month period.

BOLETO'S STABLE ~20% SHARE

is strategically significant — boleto users are disproportionately from Brazil's underbanked population (predominantly Northeast and lower-income demographics). This segment cannot easily switch to credit cards, making them sticky but also financially constrained. Any boleto discontinuation or friction increase would immediately impact ~20% of revenue. The voucher spikes in certain months (visible as green slivers) likely correspond to promotional campaigns — cross-referencing with Chart 3 could identify which categories benefited.

ACTIONABLE SIGNAL FOR SENIOR MANAGER

If credit card share drops below 70% in any month, investigate immediately — it may indicate a payment gateway issue, a shift in customer demographics, or a competitor offering better instalment terms. Conversely, if boleto share rises above 25%, the business may be acquiring more price-sensitive customers who require lower AOV products.

6

Late Delivery Rate by Product Category

% of orders delivered after promised date — top 15 worst categories

Bar Chart (sorted)

CHART TYPE

Bar Chart (sorted DESC)

DATASET

ds_order_enriched

METRIC

MAX(late_rate_pct)

Actual Data — Late Delivery Rates (from dashboard screenshot)

#	Category	Late Rate %	Risk Level
1	home_comfort_2	16.7%	CRITICAL
2	2nd category	13.5%	HIGH
3	3rd category	12.9%	HIGH
4	4th category	12.8%	HIGH
5	5th category	10.9%	MEDIUM
6	6th category	10.5%	MEDIUM
7–15	christmas_supplies and others	9.0–10.0%	WATCH

Cross-reference from **Test 1 Part B** analytics query identifies the worst categories as: furniture_mattress_and_upholstery (~14%), auto (~12%), fashion_underwear_beach (~12%), home_comfort (~11%), electronics (~10%), christmas_supplies (~10%).

Why This Chart Belongs on a Senior Manager Dashboard?

Late delivery directly impacts review scores (visible in Chart 9) and customer churn. A Senior Manager overseeing category or logistics decisions can use this chart to prioritise conversations with sellers or carrier partners. Categories with bulky/heavy items (furniture, auto accessories) have structurally higher late rates — the Senior Manager needs to know whether that's a seller-side packaging issue or a carrier-side dimensional weight problem, and this chart surfaces the "where" clearly.

KEY BUSINESS INSIGHT — BULKY PRODUCTS ARE THE LOGISTICS WEAK POINT

Categories with the worst late delivery rates share a common characteristic: large, heavy, or irregularly shaped products ([home_comfort_2](#) at 16.7%, furniture, auto accessories). These categories likely incur dimensional weight surcharges that cause carriers to deprioritise them. In contrast, standardised lightweight products (books, cds/dvds, electronics accessories) have the lowest late rates.

ACTION:

Negotiate category-specific SLAs with logistics partners, or add buffer days to delivery estimates for high-late-rate categories to reduce customer disappointment and protect review scores.

CROSS-CHART ALERT: LATE DELIVERY × REVIEW SCORE

Cross-filtering this chart with Chart 9 (Avg Review Score) will show that categories with high late delivery rates also tend to have lower review scores — confirming that delivery experience is a primary driver of customer satisfaction ratings. `security_and_services` has the lowest review score (2.50) likely related to fulfilment/service delivery issues as well.

7

Orders by Customer State

Geographic concentration — where do our customers live?

Bar Chart

CHART TYPE

Bar Chart (sorted)

DATASET

ds_order_enriched

DIMENSION

customer_state

What It Shows

A bar chart ranking all Brazilian states by total distinct order count. São Paulo (SP) is visually dominant at ~40k orders, creating a dramatic "long tail" distribution. The chart shows 27 states, making the geographic concentration immediately obvious.

~40k

SP — São Paulo

~40% of all orders

Top 3

SP + RJ + MG

~65% of all orders

24

States with <2k orders

Expansion opportunity

Why Geographic Distribution Matters for a Senior Manager?

Geographic concentration indicates both market saturation risk and expansion opportunity. If 65% of revenue comes from 3 states (SP, RJ, MG), the business is exposed to regional economic shocks in the Southeast. States in the Northeast (BA, PE, CE, MA) show low order volumes despite representing ~30% of Brazil's population — a classic market under-penetration story driven by logistics constraints (higher delivery costs, longer transit times, fewer carrier options).

KEY BUSINESS INSIGHT — SÃO PAULO DEPENDENCY RISK + NORTHEAST OPPORTUNITY

SP's ~40% order concentration creates a structural single-point risk. However, cross-referencing AOV by state typically shows that Northeast customers spend comparable amounts per order when they do purchase — suggesting the barrier is logistics access, not purchasing power.

RECOMMENDATION:

A targeted logistics investment in the Northeast (regional warehouse in Recife or Salvador) could unlock 10–15% additional order volume without requiring any change to marketing spend or product strategy.

CHART TYPE

Bar Chart

DATASET

ds_customer_segments

DIMENSION

segment

Actual Segment Data (from dashboard screenshot)

new	93.8%	92,700 customers
returning	2.7%	2,630 customers
at_risk	2.4%	2,340 customers
churned	1.2%	1,140 customers

Segment	Count	% of Total	Definition	Strategic Priority
new	92,700	93.8%	Exactly 1 order placed	Convert to returning via post-purchase email, loyalty program
returning	2,630	2.7%	2+ orders, last ≤ 180 days ago	Nurture — highest LTV segment, protect at all costs
at_risk	2,340	2.4%	2+ orders, last 181–365 days ago	Win-back campaign — still reachable, high conversion potential
churned	1,140	1.2%	2+ orders, last > 365 days ago	Low-cost reactivation only (discount voucher blast)

CRITICAL INSIGHT — 93.8% ONE-TIME BUYERS: THE RETENTION CRISIS

93.8% OF ALL CUSTOMERS (92,700 OUT OF 98,810) HAVE PLACED ONLY ONE ORDER.

This is the most important business finding in the entire dashboard. Returning customers who have re-ordered are only 2.7% of the base (2,630 people) — yet based on comparable e-commerce benchmarks, returning customers typically spend 2–3× more per order than first-time buyers due to higher trust and lower purchase hesitation. The business is functioning as a customer acquisition machine but has near-zero retention machinery in place.

IMMEDIATE ACTION: TARGET THE AT_RISK SEGMENT (2,340 CUSTOMERS)

The at_risk segment represents customers who demonstrated loyalty (purchased more than once) but haven't bought in 6–12 months. They are *within reactivation range* — still responsive to the brand but at risk of becoming churned. A targeted win-back campaign (personalised email with 15% discount, product recommendation based on prior purchase history) targeting these 2,340 customers is the

HIGHEST-ROI RETENTION INVESTMENT

available right now.

Average Review Score by Product Category

Customer satisfaction signal — which categories disappoint?

Bar Chart (sorted)

+ ADDED

CHART TYPE

Bar Chart (sorted ASC)

DATASET

ds_order_enriched

METRIC

AVG(avg_review_score)

What It Shows

A bar chart showing the average review score per product category (scale 1–5), sorted ascending to surface the worst-performing categories first. The chart uses the `avg_review_score` field from `ds_order_enriched`, which is pre-computed as `ROUND(AVG(r.review_score), 1)` per order. Superset then takes `AVG(avg_review_score)` per category across all orders.

Actual Data — Review Scores by Category (from dashboard screenshot)

Category	Avg Review Score	Assessment
security_and_services	2.50	CRITICAL — Far below platform average
furniture_mattress_and_upholstery	3.62	BELOW AVG — Likely delivery-related
Several mid-range categories	3.73 – 3.84	AVERAGE — Monitor
home_comfort (highest visible)	3.87	GOOD — Benchmark target

Platform-wide average review score across all categories is approximately **3.8 – 3.9 / 5**.

Why Include Review Score on a Senior Manager Dashboard?

Review scores are a **leading indicator** of future churn and retention. A category with a falling review score will see declining repeat purchases before the revenue impact is visible in the Revenue Trend chart (Chart 2). The Senior Manager can use this chart proactively to investigate and intervene — before the problem appears in monthly revenue numbers. It also enables the critical cross-filter connection with Chart 6 (Late Delivery) to test the hypothesis that delivery delays cause bad reviews.

KEY BUSINESS INSIGHT — SECURITY_AND_SERVICES IS AN OUTLIER PROBLEM

`security_and_services` scores

2.50/5

— nearly 1.3 points below the platform average. This is not noise: it likely reflects a structural product or service quality issue in this category (installation complexity, service reliability, or post-purchase support gaps). While this category may not be in the top 10 by revenue, its review score depresses the platform-level rating.

ACTION:

The Senior Manager should initiate a seller audit for `security_and_services` category and consider whether these products meet the platform's quality standards for continued listing.

CROSS-CHART ANALYSIS: REVIEW SCORE × LATE DELIVERY

Cross-filtering Chart 9 and Chart 6 confirms the delivery–satisfaction link: categories with late delivery rates above 12% (home_comfort_2, furniture categories) also show review scores below 3.7. The inverse is also true — categories with low late rates (books, standard electronics) tend to score ≥ 3.85 . This validates the operational investment case for improving delivery SLAs in high-late-rate categories.

13 Integrated Business Narrative — Strategic Insights

What the dashboard tells us when all 9 charts are read together

Reading each chart in isolation misses the connections. The following narrative synthesises all 9 charts into a coherent business story for the Senior Manager.

The Story in Four Acts

Act 1 — Hyper-growth, then plateau (Charts 1 & 2)

The business grew 8.6× from Jan–Nov 2017, peaking at R\$ 1.19M in November (Black Friday). Since then, monthly revenue has plateaued at ~R\$ 1M. The platform has reached a growth ceiling. The KPI cards confirm this is not a temporary dip — it is structural. **Root cause hypothesis: customer acquisition has slowed and retention is failing (see Act 3).**

Act 2 — The product portfolio is concentrated but healthy at the top (Charts 3 & 9)

The top 10 categories account for the majority of revenue. health_beauty leads in volume (R\$1.3M, driven by repeat-purchase behaviour), while watches_gifts leads in value (R\$254 AOV — a premium segment). Review scores are generally acceptable (3.7–3.87) across most categories, with one critical outlier: security_and_services at 2.50/5. **The category mix is healthy — the bottleneck is retention, not product-market fit.**

Act 3 — The retention crisis is the central strategic problem (Chart 8)

93.8% of all customers have placed exactly one order. Only 2.7% have returned (2,630 customers). This is the most important insight in the dashboard. In a healthy e-commerce business, repeat purchases account for 30–50% of revenue. Here, the business is almost entirely dependent on first-time buyers. Every R\$ spent on customer acquisition is lost if the customer does not return. The 2,340 at_risk customers (purchased 2+ times, last 6–12 months ago) are the highest-priority win-back target.

Act 4 — Operational risks are specific and addressable (Charts 4, 6 & 7)

96%+ of orders are delivered (Chart 4 — healthy pipeline). However, specific categories have late delivery rates of 13–17% (Chart 6), which directly depress review scores (Chart 9) and contribute to churn. Geographic concentration in SP/RJ/MG (Chart 7) creates both saturation risk and an expansion opportunity in the underserved Northeast.

Top 3 Recommendations for the Senior Manager

Priority	Action	Metric to Watch	Expected Impact
P1 — Urgent	Launch retention program: Win-back email for at_risk segment (2,340 customers) + post-purchase loyalty incentives for new customers to trigger second order	Returning % in Chart 8; Revenue per customer in Chart 1	Even 10% conversion of at_risk → returning generates ~234 returning customers, potentially R\$75–100k additional revenue at 2× AOV
P2 — Short-term	Fix logistics for high-late-rate categories: Negotiate SLA with carriers for furniture/auto categories; add 2-3 buffer days to delivery	Late delivery rate in Chart 6; Avg review score in Chart 9	Reducing home_comfort_2 late rate from 16.7% to 10% would improve review score, reduce

estimates for home_comfort_2 and similar

support contacts, and protect repeat purchases in that category

P3 —
Strategic

Northeast expansion: Pilot 1 regional fulfilment point (Recife or Fortaleza) to service BA/PE/CE/MA states; target Boleto-friendly product categories (consumables, health_beauty)

Orders by State in Chart 7; Boleto share in Chart 5

Northeast represents ~30% of Brazil's population with currently <5% order share — addressable market with existing platform infrastructure

14 Bonus Points Summary

All three bonus requirements implemented and verified in the live Superset dashboard

Bonus Requirement	Implementation	Evidence
★ 1. MoM Revenue KPI Card Big Number showing % change vs previous month	Charts 1A/1B/1C use Superset's <i>Big Number w/ Trendline</i> chart type with <i>Time Comparison</i> set to "1 month ago". Shows current total + coloured arrow + MoM % automatically. No custom SQL needed — Superset's temporal engine handles the comparison.	Visible in Dashboard Screenshot 1 — "KPI - Total Revenue (MoM)" top-left card
★ 2. Cross-Filtering Clicking one chart filters all others on the dashboard	Enabled via Dashboard → Edit → Enable Cross-Filtering toggle. Any click on a category bar (Chart 3), payment type (Chart 5), state bar (Chart 7), or segment (Chart 8) propagates a filter to all compatible charts simultaneously. No additional configuration per chart required after dashboard-level activation.	Dashboard-level setting — testable by clicking any chart element in Superset
★ 3. Custom SQL Dataset (3+ Table JOIN) At least one dataset joining ≥ 3 tables	Two datasets created in SQL Lab: <code>ds_order_enriched</code> : 7-table JOIN — orders + customers + order_items + products + translation + order_payments + order_reviews. <code>ds_customer_segments</code> : 3-table JOIN — orders + customers + order_payments. Both promoted to virtual datasets reused across 8 of 9 charts.	SQL Lab → Saved Queries → Promoted to Datasets. Shown in full in Section 3

Complete Chart Summary

#	Chart Title	Type	Primary Business Question Answered	Bonus
1	KPI Scorecard (Revenue, Orders, AOV)	Big Number ×3	Is the business growing or declining this month?	★ MoM
2	Monthly Revenue Trend	Line Chart	What is the revenue trajectory over 24 months?	
3	Top 10 Categories by Revenue	Bar Chart	Which product categories drive the most revenue?	
4	Order Status Distribution	Donut Chart	What % of orders are successfully delivered?	
5	Revenue by Payment Method	Stacked Bar	How does payment mix evolve and what are the risks?	

6	Late Delivery Rate by Category	Bar Chart	Which categories have logistics problems?
7	Orders by Customer State	Bar Chart	Where is our customer base concentrated?
8	Customer Segment Distribution	Bar Chart	How healthy is our retention funnel?
9	Avg Review Score by Category	Bar Chart	Which categories are disappointing customers?
Cross-filtering: all charts			★ XFilter
ds_order_enriched: 7-table JOIN			★ JOIN

★ ALL THREE BONUS POINTS IMPLEMENTED

(1) Big Number MoM comparison cards for all three KPIs using Superset's native Time Comparison engine. (2) Cross-filtering enabled at dashboard level — clicking any chart element filters all compatible charts. (3) Two custom SQL datasets created in SQL Lab: `ds_order_enriched` (7-table JOIN) and `ds_customer_segments` (3-table JOIN), both promoted to virtual datasets.

Recommended Walkthrough Order for Interview Presentation

SUGGESTED NARRATIVE FLOW

1. KPI CARDS (CHART 1)

— Set the stage: "Here's where the business is today"

2. REVENUE TREND (CHART 2)

— Tell the growth story: 8.6× in 11 months, then plateau

3. CATEGORIES (CHART 3)

— Show cross-filtering: click "health_beauty" to drill down

4. CUSTOMER SEGMENTS (CHART 8)

— Land the strategic alarm: 93.8% one-time buyers

5. LATE DELIVERY (CHART 6)

— Operational risk: 16.7% late rate in home_comfort_2

6. REVIEW SCORE (CHART 9)

— Connect the dots: bad delivery → bad reviews

7. STATE DISTRIBUTION (CHART 7)

— Close with opportunity: Northeast expansion

